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Cuff stitching reinforcement

Abstract

A prosthetic heart valve includes a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts. A cuff may be coupled to the stent so that a flat, bottom edge of the cuff lies adjacent the proximal end of the stent. A pattern of stitches may be circumferentially disposed around the flat bottom edge of the cuff, the pattern of stitches alternating between stitches sewn to the cuff only and stitches sewn to both the cuff and the stent.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) The present application is a divisional of U.S. patent application Ser. No. 16/397,234, filed Apr. 29, 2019, which is a continuation of U.S. patent application Ser. No. 14/548,723, filed Nov. 20, 2014, now U.S. Pat. No. 10,314,693, which claims the benefit of the filing date of U.S. Provisional Patent Application No. 61/909,496, filed Nov. 27, 2013, the disclosure of which is hereby incorporated herein by reference.

BACKGROUND OF THE INVENTION

(1) The present invention relates to heart valve replacement and, in particular, to collapsible prosthetic heart valves. More particularly, the present invention relates to collapsible prosthetic heart valves having improved cuff attachments.

(2) Prosthetic heart valves that are collapsible to a relatively small circumferential size can be delivered into a patient less invasively than valves that are not collapsible. For example, a collapsible valve may be delivered into a patient via a tube-like delivery apparatus such as a catheter, a trocar, a laparoscopic instrument, or the like. This collapsibility can avoid the need for a more invasive procedure such as full open-chest, open-heart surgery.

(3) Collapsible prosthetic heart valves typically take the form of a valve structure mounted on a stent. There are two types of stents on which the valve structures are ordinarily mounted: a self-expanding stent and a balloon-expandable stent. To place such valves into a delivery apparatus and ultimately into a patient, the valve must first be collapsed or crimped to reduce its circumferential size.

(4) When a collapsed prosthetic valve has reached the desired implant site in the patient (e.g., at or near the annulus of the patient's heart valve that is to be replaced by the prosthetic valve), the prosthetic valve can be deployed or released from the delivery apparatus and re-expanded to full operating size. For balloon-expandable valves, this generally involves releasing the valve, assuring its proper location, and then expanding a balloon positioned within the valve stent. For self-expanding valves, on the other hand, the stent automatically expands as the sheath covering the valve is withdrawn.

SUMMARY OF THE INVENTION

(5) In some embodiments, a prosthetic heart valve includes a collapsible and expandable stent

having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts. A cuff may be coupled to the stent so that a flat, bottom edge of the cuff lies adjacent the proximal end of the stent. A pattern of stitches may be circumferentially disposed around the flat bottom edge of the cuff, the pattern of stitches alternating between stitches sewn to the cuff only and stitches sewn to both the cuff and the stent.

(6) In some embodiments, a prosthetic heart valve may include a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts and a plurality of commissure features. A cuff may be coupled to the stent so that a top edge of the cuff lies adjacent the plurality of commissure features and a plurality of tethers incorporated along the top edge of the cuff and coupled to the cuff only.

(7) In some embodiments, a method of making a prosthetic heart valve may include (i) providing a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts, (ii) coupling a cuff to the stent so that a flat bottom edge of the cuff lies adjacent the proximal end of the stent, and (iii) sewing a pattern of stitches circumferentially around the flat bottom edge of the cuff, the pattern of stitches alternating between stitches sewn to the cuff only and stitches sewn to both the cuff and the stent.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments of the presently disclosed heart valves are disclosed herein with reference to the drawings, wherein:

(2) FIG. 1 is a side elevational view of a conventional prosthetic heart valve;

(3) FIG. 2 is a highly schematic cross-sectional view showing another prosthetic heart valve disposed within a native valve annulus;

(4) FIG. 3A is a developed view of a cuff having a series of peaks;

(5) FIG. 3B is a perspective side view of the cuff of FIG. 3A after the attachment portions of the cuff have been coupled together;

(6) FIG. 4A is a partial schematic side view of a first variation of a prosthetic heart valve having a cuff coupled to a stent, the cuff having a plurality of cutouts;

(7) FIG. 4B is a partial schematic side view of a second variation of a prosthetic heart valve having a cuff coupled to a stent, the cuff having a substantially straight inflow edge;

(8) FIGS. 4C-4D are partial schematic side views of examples of cuffs having substantially straight inflow edges;

(9) FIG. 4E is a highly schematic end view of the prosthetic heart valve of FIG. 4B disposed within a native valve annulus;

(10) FIG. 5 is a developed view of a cuff having stitches attached near the inflow end;

(11) FIG. 6 is a highly schematic end view of the prosthetic heart valve of FIG. 5 disposed within a native valve annulus;

(12) FIGS. 7A and 7B are a highly schematic end views of a prosthetic heart valve disposed within a generally elliptical native valve annulus, the prosthetic heart valve having and not having cuff inflow end stitches, respectively; and

(13) FIG. 8 is a developed view of a cuff having stitches attached near the outflow edge.

(14) Various embodiments of the present invention will now be described with reference to the appended drawings. It is to be appreciated that these drawings depict only some embodiments of the invention and are therefore not to be considered limiting of its scope.

DETAILED DESCRIPTION OF THE INVENTION

(15) Inaccurate deployment and anchoring of a prosthetic heart valve may result in the leakage of blood between the implanted heart valve and the native valve annulus, commonly referred to as perivalvular (or “paravalvular”) leakage. In aortic valves, this leakage enables blood to flow from the aorta back into the left ventricle, reducing cardiac efficiency and putting a greater strain on the heart muscle. Additionally, calcification of the aortic valve may affect performance and the interaction between the implanted valve and the calcified tissue is believed to be relevant to leakage, particularly when the native valve leaflets have not first been resected. To reduce the risk of leakage, adequate anchoring and sealing are helpful. Moreover, anatomical variations from one patient to another may affect wear and durability.

(16) As used herein, the term “proximal,” when used in connection with a prosthetic heart valve in the aortic position, refers to the end of the heart valve closest to the heart when the heart valve is implanted in a patient, whereas the term “distal,” when used in connection with a prosthetic heart valve, refers to the end of the heart valve farthest from the heart when the heart valve is implanted in a patient. Also, as used herein, the word “about” is intended to mean that slight variations from absolute are included within the scope of the valve recited, for example, due to manufacturing tolerances.

(17) FIG. 1 shows a collapsible stent-supported prosthetic heart valve **100** including a stent **102** and a valve assembly **104** as is known in the art. The prosthetic heart valve **100** is designed to replace the native tricuspid valve of a patient, such as the native aortic valve. It should be noted that while the inventions herein are described predominantly in connection with their use with a prosthetic aortic valve and a stent having a shape as illustrated in FIG. 1, the valve could be a bicuspid valve, such as the mitral valve, and the stent could have different shapes, such as a flared or conical annulus section, a less-bulbous aortic section, and the like, and a differently shaped transition section.

(18) Prosthetic heart valve **100** will be described in more detail with reference to FIG. 1. Prosthetic heart valve **100** includes expandable stent **102** which may be formed from biocompatible materials that are capable of self-expansion, such as, for example, shape memory alloys, such as the nickel-titanium alloy known as “Nitinol” or other suitable metals or polymers. Stent **102** extends from proximal or annulus end **130** to distal or aortic end **132**, and includes annulus section **140** adjacent proximal end **130**, transition section **141**, and aortic section **142** adjacent distal end **132**. Annulus section **140** has a relatively small cross-section in the expanded condition, while aortic section **142** has a relatively large cross-section in the expanded condition. Preferably, annulus section **140** is in the form of a cylinder having a substantially constant diameter along its length. Transition section **141** may taper outwardly from annulus section **140** to aortic section **142**. Each of the sections of stent **102** includes a plurality of struts **160** forming cells **162** connected to one another in one or more annular rows around the stent. For example, as shown in FIG. 1, annulus section **140** may have two annular rows of complete cells **162** and aortic section **142** and transition section **141** may each have one or more annular rows of partial cells **162**. Cells **162** in aortic section **142** may be larger than cells **162** in annulus section **140**. The larger cells in aortic section **142** better enable prosthetic valve **100** to be positioned in the native valve annulus without the stent structure interfering with blood flow to the coronary arteries.

(19) Stent **102** may include one or more retaining elements **168** at distal end **132** thereof, retaining elements **168** being sized and shaped to cooperate with female retaining structures (not shown) provided on a deployment device configured to deploy the prosthetic valve **100** in the native valve annulus of a patient. The engagement of retaining elements **168** with the female retaining structures on the deployment device helps maintain prosthetic heart valve **100** in assembled relationship with the deployment device, minimizes longitudinal movement of the prosthetic heart valve relative to the deployment device during unsheathing or resheathing procedures, and helps prevent rotation of the prosthetic heart valve relative to the deployment device as the deployment device is advanced

to the target location and the heart valve deployed.

(20) Prosthetic heart valve **100** includes valve assembly **104**, preferably positioned in annulus section **140** of the stent **102** and secured to the stent. Valve assembly **104** includes cuff **176** and a plurality of leaflets **178** which collectively function as a one-way valve by coapting with one another. As a prosthetic aortic valve, valve **100** has three leaflets **178**. However, it will be appreciated that other prosthetic heart valves with which the sealing portions of the present disclosure may be used may have a greater or lesser number of leaflets **178**.

(21) Although cuff **176** is shown in FIG. **1** as being disposed on the luminal or inner surface of annulus section **140**, it is contemplated that cuff **176** may be disposed on the abluminal or outer surface of annulus section **140** or may cover all or part of either or both of the luminal and abluminal surfaces. Both cuff **176** and leaflets **178** may be wholly or partly formed of any suitable biological material or polymer such as, for example, polytetrafluoroethylene (PTFE), ultra-high molecular weight polyethylene, polyurethane, polyvinyl alcohol, silicone or combinations thereof.

(22) Leaflets **178** may be attached along their belly portions to cells **162** of stent **102**, with the commissure between adjacent leaflets **178** attached to commissure features **166** of the stent. As can be seen in FIG. **1**, each commissure feature **166** may lie at the intersection of four cells **162**, two of the cells being adjacent one another in the same annular row, and the other two cells being in different annular rows and lying in end-to-end relationship. Preferably, commissure features **166** are positioned entirely within annulus section **140** or at the juncture of annulus section **140** of stent **102** and transition section **141**. Commissure features **166** may include one or more eyelets which facilitate the suturing of the leaflet commissure to stent **102**.

(23) Prosthetic heart valve **100** may be used to replace a native aortic valve, a surgical heart valve or a heart valve that has undergone a surgical procedure. Prosthetic heart valve **100** may be delivered to the desired site (e.g., near the native aortic annulus) using any suitable delivery device. During delivery, prosthetic heart valve **100** is disposed inside the delivery device in the collapsed condition. The delivery device may be introduced into a patient using a transfemoral, transapical, transseptal or any other percutaneous approach. Once the delivery device has reached the target site, the user may deploy prosthetic heart valve **100**. Upon deployment, prosthetic heart valve **100** expands so that annulus section **140** is in secure engagement within the native aortic annulus. When prosthetic heart valve **100** is properly positioned inside the heart, it works as a one-way valve, allowing blood to flow from the left ventricle of the heart to the aorta, and preventing blood from flowing in the opposite direction.

(24) FIG. **2** is a highly schematic cross-sectional illustration of prosthetic heart valve **200** disposed within native valve annulus **250**. As seen in the figure, stent **202** has a substantially circular cross-section which is disposed within the non-circular native valve annulus **250**. At certain locations around the perimeter of heart valve **200**, crescent-shaped gaps **240** form between the heart valve and native valve annulus **250**. Blood flowing through these gaps and past valve assembly **204** of prosthetic heart valve **200** may be undesirable.

(25) FIGS. **3A** and **3B** illustrate the outer surface of a cuff before coupling to a stent. As shown, cuff **300** includes a generally parallelogram-shaped body **302**, a series of trapezoidal or triangular peaks **304a**, **304b**, and **304c** and a pair of attachment portions **306**. It will be understood, however, that cuff **300** may be formed in various geometric shapes and that illustrated configurations are merely exemplary. Peaks **304** may be used to couple cuff **300** to a stent. Body **302** extends longitudinally from an inflow end **320** to an outflow end **322**, peaks **304a**, **304b**, and **304c** being located near outflow end **322**. Attachment portions **306** are adapted to be coupled together to form cuff **300** into the wrapped or assembled configuration shown in FIG. **3B**. Attachment portions **306** may overlap one another in the wrapped configuration, and may be coupled together using a suture, an adhesive or any other suitable means. Attachment portions **306** may also abut one another and couple together. Cuff **300** may be placed in the wrapped configuration before being coupled to a stent. Alternatively, cuff **300** may be first coupled to the stent after which attachment portions **306**

may be coupled together.

(26) Cuff **300** is coupled to a stent and to a number of leaflets to form a prosthetic heart valve. It will be understood that the components may be assembled using various techniques and in different orders. For example, the cuff may be coupled to the stent followed by the leaflets or the leaflets may be attached to the cuff followed by assembly of both to the stent.

(27) Two variations of the final cuff-stent assembly are described below with reference to FIGS. **4A-C**. In a first variation, shown in FIG. **4A**, prosthetic heart valve **400A** includes stent **410A** formed of a plurality of struts **412A**, with cuff **420A** coupled to the stent. Cuff **420A** includes body **422A** extending between inflow end **430A** and outflow end **432A**.

(28) Excess portions of cuff **420A** may unduly increase the crimp profile of heart valve **400A** and may also impinge on the effective orifice area (e.g., the entrance orifice through which blood flows to the valve assembly). In order to address these concerns, excess portions **440A** of body **422A** near inflow end **430A** may be trimmed using a cutting mandrel, a die or other suitable means. A fixation device may be useful in this trimming process. One fixation device useful for this purpose is shown in U.S. Provisional Patent Application Ser. No. 61/666,174 entitled "VALVE ASSEMBLY FOR CRIMP PROFILE" filed Jun. 29, 2012, the content of which is hereby incorporated by reference herein in its entirety.

(29) The trimming of cuff **420A** may be accomplished either prior to or after the attachment of cuff **420A** to stent **410A**. After completion, cutouts **424A**, such as the semicircular cutouts shown in FIG. **4A**, are formed in cuff **420A** at recurring positions along the edge near inflow end **430A**. In this variation, cutouts **424A** are configured to mitigate effective orifice area impingement of cuff **420A** during forward flow of blood through inflow end **430A**. Though cutouts **424A** may reduce the crimp profile of the device, in applications where prosthetic heart valve **400A** is implanted in an asymmetric annulus or in an annulus having heavily calcified regions as described above with reference to FIG. **2**, cutouts **424A** may not adequately seal inflow end **430A** against the surrounding tissue.

(30) In a second variation, shown in FIG. **4B**, prosthetic heart valve **400B** includes stent **410B** formed of a plurality of struts **412B**, with cuff **420B** coupled to the stent. Unlike heart valve **400A**, cuff **420B** of heart valve **400B** is not trimmed along either inflow end **430B** or outflow end **432B**, leaving a substantially flat edge **424B** near inflow end **430B**. Flat edge **424B** may align with the proximal ends of struts **412** near inflow end **430B**. Flat edge **424B** may present some drawbacks as explained below.

(31) FIGS. **4C-4D** are partial schematic side views of examples of cuffs having untrimmed inflow edges. In FIG. **4C**, prosthetic heart valve **400C** includes stent **410C** formed of a plurality of struts **412C**, with cuff **420C** coupled to the stent. Cuff **420C** is rolled on itself near inflow end **430C** to create halo **436C**, a ring-like portion of cuff **420C** that extends past flat edge **424C**. Halo **436C** may be formed of the same material as cuff **420C** or a different material. Additionally, halo **436C** may be integrally formed with cuff **420C** or may be formed of a different portion of tissue, fabric or other cuff material that is coupled to cuff **420C** through adhesive, suture or other suitable techniques. In FIG. **4D**, prosthetic heart valve **400D** includes stent **410D** formed of a plurality of struts **412D**, with cuff **420D** coupled to the stent. Cuff **420D** is rolled on itself near inflow end **430D** to create parachute **436D**.

(32) FIG. **4E** is a highly schematic end view illustrating prosthetic heart valve **400B** of FIG. **4B** disposed within native valve annulus **250** as seen through inflow end **430B**. When cutouts, such as those shown in FIG. **4A**, are not formed in cuff **420B**, portions of the cuff may bend inwardly during forward flow of blood. Specifically, bent portions **440** of cuff **420B** that are not directly attached to struts **412B** of stent **410B** fold inwardly toward the central axis of prosthetic heart valve **400B**, diminishing the effective orifice area **450** through which blood may flow. In some experiments, it has been found that bent portions **440** may diminish the total effective orifice area **450** by as much as about 10% to about 20%. Moreover, blood flowing through regions near bent

portions **440** may be directed outside of cuff **420B** as opposed to flowing through orifice area **450** of the valve assembly, leading to perivalvular leakage. Thus, bent portions **440** may cause regurgitation and other inefficiencies which reduce cardiac performance. Similar issues may also arise in examples having untrimmed portions as shown in FIGS. **4C** and **4D**.

(33) Several techniques will be described to reduce the formation of bent portions in the cuff and to avoid constriction of the effective orifice area through the prosthetic heart valve. FIG. **5** illustrates one such technique, which uses stitches to reduce bent portions of the cuff. For the sake of illustration, cuff **500** is shown flat, as if stretched out and only a portion of a stent is shown in detail. Cuff **500** includes body **502**, a series of large peaks **504a**, **504b**, and **504c** projecting from one edge of body **502**, and a pair of attachment portions **506**, one on each longitudinal end of body **502**. To minimize the formation of bent portions of the cuff during use, as shown in FIG. **4E**, a stitching technique may be applied along flat edge **524** of body **502** near inflow end **530**. In one example, a stitch pattern P may be disposed along flat edge **524** to eliminate or minimize bending of the cuff.

(34) An enlarged detailed view of FIG. **5** is shown of body **502** as well as portions of select struts **552a**, **552b** (shown in broken lines) to which body **502** will be attached. For the sake of clarity, the remaining portions of the stent are not shown. As shown in the detailed view, when two struts **552a** and **552b** join at the inflow end of stent **550**, a horseshoe **554** is formed. Stitch pattern P may include a first horseshoe locking stitch L1 anchored at each horseshoe **554**. Between adjacent horseshoe locking stitches L1, stitch pattern P may include additional locking stitches L2 sewn to cuff **500** only, as shown in the enlarged detailed view in FIG. **5**. Each locking stitch may include two threads that entwine together at the cuff. Although FIG. **5** shows two such locking stitches L2 between adjacent horseshoe locking stitches L1, cuff **500** may include a greater or lesser number of locking stitches L2 as desired. To get from one horseshoe locking stitches L1 to another, any type of stitch may be used though locking stitches L2 are shown which minimize bulkiness. Thus, stitch pattern P may begin at the attachment portion **506** at one end of cuff **500** and form a recurring pattern of a single horseshoe locking stitch L1 coupling cuff **500** to a horseshoe **554** of stent **550** and two additional locking stitches L2 sewn to the cuff only. Stitch pattern P may begin and end in the attachment portions **506** so that the beginning and end knots are hidden when the cuff **500** is fully attached to stent **550** and the attachment portions are joined together.

(35) Such a stitching technique allows the prosthetic heart valve to achieve a small crimp profile when collapsed for delivery, while securing each section of cuff **500** to stent **550** and minimizing bending of cuff tissue into the flow area. Additionally, it is common to create a stent **550** having a fully-expanded diameter that is greater than the diameter the stent will have in use. Thus, stitch pattern P may have the added benefit of ensuring optimal valve performance by limiting cuff **500** from over-expanding outside the optimal use range. Finally, stitch pattern P may reduce or eliminate the inward bending of cuff **500** at inflow end **530**, thereby improving sealing by providing a larger landing zone and maintaining the effective orifice area through which blood may flow, while at the same time increasing the landing area along which the native valve annulus may be positioned relative to the prosthetic heart valve allowing easier placement by the physician. Larger landing areas may provide more forgiving placement accuracy. Stitch pattern P may provide these benefits while not unduly increasing the crimp profile of the prosthetic heart valve.

(36) FIG. **6** is a highly schematic end view illustrating prosthetic heart valve **600** disposed within native valve annulus **250**. Prosthetic heart valve **600** includes cuff **620** coupled to stent **610**. A stitching technique as described with reference to FIG. **5** has been applied to cuff **620**. Specifically, stitch pattern P has been applied along the circumference of cuff **620** at its inflow end. Because of the stitching technique, bent portions **630** of the cuff are reduced, leading to a larger effective orifice area **650** through which blood may flow and more adequate sealing with surrounding tissue at the inflow end of prosthetic heart valve **600**.

(37) The advantages of the stitching technique described above will also be readily seen when a

prosthetic heart valve is implanted in an irregularly-shaped or elliptical native valve annulus. FIGS. 7A and 7B are highly schematic cross-sectional illustrations of prosthetic heart valves **700A** and **700B** disposed within an elliptical native valve annulus **250**. Prosthetic heart valve **700A** includes cuff **720A** coupled to stent **710A**, but does not employ the stitching technique described above. When prosthetic heart valve **700A** is deployed in elliptical native annulus **250**, bicuspid or mitral valves and subjected to blood flow, bent portions **730A** are formed, reducing the effective orifice area **750A** through which blood may flow. Prosthetic heart valve **700B** is similar to heart valve **700A**, and includes cuff **720B** coupled to stent **710B**. However, in prosthetic heart valve **700B**, the stitching technique described above with reference to FIG. 5 has been applied to cuff **720B**. When prosthetic heart valve **700B** is deployed in elliptical native annulus **250** and subjected to blood flow, the formation of bent portions **730B** is minimized, leading to great apposition of cuff **720B** to stent **710B**, a larger effective orifice area **750B** through which blood may flow, and more adequate sealing with surrounding tissue at the inflow end of the prosthetic heart valve.

(38) FIG. 8 illustrates another example of a cuff for promoting the sealing of the prosthetic heart valve against surrounding tissue and maximizing the effective orifice area through which blood may flow through the prosthetic heart valve. Cuff **800** includes body **802**, a series of peaks **804a**, **804b**, and **804c** at the outflow end **832** of body **802**, and a pair of attachment portions **806**, one on each longitudinal end of body **802**. Body **802** extends between inflow end **830** and outflow end **832**. To minimize the formation of bent portions of the cuff during use, as shown in FIG. 4E, a tethering technique may be applied near outflow end **832** of cuff **800**. In one example, a plurality of horizontal tethers **T1** may be used to attach peaks **804a**, **804b**, and **804c** to one another. Though cuff **800** is shown in the flat configuration, it will be understood that when the cuff is in the wrapped configuration, one set of tethers **T1** attach peak **804a** to peak **804c**. Vertical tethers **T2** attach the horizontal tethers **T1** to one another, creating a net-like structure or web. By adding tethers **T1** and **T2** near outflow end **832** of cuff **800**, the landing zone of the prosthetic heart valve is increased without unduly increasing its crimp profile. In general, a longer landing zone corresponds to better sealing. In some variations, instead of using separate horizontal tethers **T1** and vertical tethers **T2**, a preformed webbing, netting or mesh may be attached to the top of body **802** or peaks **804a**, **804b**, **804c** to increase the landing zone in a similar manner.

(39) Although the invention herein has been described with reference to particular embodiments, it is to be understood that these embodiments are merely illustrative of the principles and applications of the present invention. It is therefore to be understood that numerous modifications may be made to the illustrative embodiments and that other arrangements may be devised without departing from the spirit and scope of the present invention as defined by the appended claims.

(40) In some embodiments, a prosthetic heart valve, includes a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts. A cuff may be coupled to the stent so that a flat, bottom edge of the cuff lies adjacent the proximal end of the stent. A pattern of stitches may be circumferentially disposed around the flat bottom edge of the cuff, the pattern of stitches alternating between stitches sewn to the cuff only and stitches sewn to both the cuff and the stent.

(41) In some examples, the pattern of stitches may include a plurality of locking stitches. The pattern of stitches may include a recurring pattern comprised of a first type of stitch and a second type of stitch. The first type of stitch may be a locking stitch sewn to the cuff only. The second type of stitch may be a locking stitch sewn to the cuff and at least a portion of the stent. The recurring pattern may include two adjacent stitches of the first type followed by a single stitch of the second type. The plurality of struts may form horseshoe-shaped portions at the proximal end of the stent, and the pattern of stitches may couple the cuff to the horseshoe-shaped portions. The pattern of stitches may include locking stitches sewn only to the cuff between the horseshoe-shaped portions.

(42) In some embodiments, a prosthetic heart valve may include a collapsible and expandable stent

having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts and a plurality of commissure features. A cuff may be coupled to the stent so that a top edge of the cuff lies adjacent the plurality of commissure features and a plurality of tethers incorporated along the top edge of the cuff and coupled to the cuff only.

(43) In some examples, the stent may include a plurality of commissure features and the top edge of the cuff includes a plurality of peaks coupled to the commissure features. The plurality of tethers may include at least one vertical tether and at least one horizontal tether. Adjacent ones of the plurality of peaks may be coupled to one another via the at least one horizontal tether. The at least one vertical tether may be coupled to the at least one horizontal tether. The at least one vertical tether may include multiple vertical tethers disposed between the plurality of peaks.

(44) In other embodiments, a method of making a prosthetic heart valve may include (i) providing a collapsible and expandable stent having a proximal end, a distal end, an annulus section adjacent the proximal end and an aortic section adjacent the distal end, the stent including a plurality of struts, (ii) coupling a cuff to the stent so that a flat bottom edge of the cuff lies adjacent the proximal end of the stent, and (iii) sewing a pattern of stitches circumferentially around the flat bottom edge of the cuff, the pattern of stitches alternating between stitches sewn to the cuff only and stitches sewn to both the cuff and the stent.

(45) In some examples, sewing a pattern of stitches may include forming a plurality of locking stitches. Sewing a pattern of stitches may include sewing a recurring pattern comprised of a first type of stitch and a second type of stitch. Sewing a pattern of stitches may include sewing a first type of stitch that is a locking stitch sewn to the cuff only. Sewing a pattern of stitches may include sewing a second type of stitch that is a locking stitch sewn to the cuff and at least a portion of the stent. The plurality of struts may form horseshoe-shaped portions at the proximal end of the stent, and sewing a pattern of stitches may include sewing the cuff to the horseshoe-shaped portions.

(46) It will be appreciated that the various dependent claims and the features set forth therein can be combined in different ways than presented in the initial claims. It will also be appreciated that the features described in connection with individual embodiments may be shared with others of the described embodiments.

Claims

1. A prosthetic heart valve, comprising: a collapsible and expandable stent having an inflow end at a proximal end, an outflow end at a distal end, an annulus section adjacent the inflow end and an aortic section adjacent the outflow end, the stent including a plurality of struts forming a plurality of cells disposed in rows; a cuff coupled to the stent so that a flat bottom edge of the cuff lies adjacent the inflow end of the stent, the cuff fully covering each cell of a proximal-most row of cells closest to the inflow end of the stent; and a recurring pattern of stitches circumferentially disposed around the flat bottom edge of the cuff and coupling the cuff to the stent only, the recurring pattern of stitches including stitches of two different types, each of the two different types being sewn to the flat bottom edge of the cuff; and wherein the two different types of stitches include a first type of stitch sewn to the cuff only.

2. The prosthetic heart valve of claim 1, wherein the two different types of stitches include a second type of stitch sewn to the cuff and to at least a portion of the stent.

3. The prosthetic heart valve of claim 1, wherein the pattern includes two adjacent stitches of a first type followed by a single stitch of a second type.

4. The prosthetic heart valve of claim 1, wherein the plurality of struts form horseshoe-shaped ends at the inflow end of the stent, and the recurring pattern of stitches couples the cuff to the horseshoe-shaped ends.

5. The prosthetic heart valve of claim 4, wherein the recurring pattern of stitches includes locking

stitches sewn only to the cuff between the horseshoe-shaped ends.

6. The prosthetic heart valve of claim 2, wherein each of the plurality of cells has apices, and each of the second type of stitch is coupled to proximal-most ones of the apices.

7. The prosthetic heart valve of claim 2, wherein each of the second type of stitch is coupled to a single location of each of a number of the plurality of cells.

8. A method of making a prosthetic heart valve, comprising: providing a collapsible and expandable stent having an inflow end at a proximal end, an outflow end at a distal end, an annulus section adjacent the inflow end and an aortic section adjacent the outflow end, the stent including a plurality of struts forming a plurality of cells disposed in rows; coupling a cuff to the stent so that a flat bottom edge of the cuff lies adjacent the inflow end of the stent so that the cuff fully covers each cell of a proximal-most row of cells closest to the inflow end of the stent; and sewing a recurring pattern of stitches circumferentially around the flat bottom edge of the cuff so as to couple the cuff to the stent only, the recurring pattern of stitches including stitches of two types; and wherein sewing a pattern of stitches comprises sewing a first type of stitch that is a locking stitch sewn to the cuff only.

9. The method of claim 8, wherein sewing the pattern of stitches comprises forming a plurality of locking stitches.

10. The method of claim 8, wherein sewing a pattern of stitches comprises sewing a recurring pattern comprised of the first type of stitch and the second type of stitch.

11. The method of claim 8, wherein sewing a pattern of stitches comprises sewing a second type of stitch sewn to the cuff and to at least a portion of the stent.
